Z-test and t-test

Z-test

- A Z-test is a statistical test that uses the standard normal (Z) distribution.
- We generally use a Z-test when the population standard deviation sigma is known.
- Population standard deviation is known.
- The sample is randomly selected, and the relevant assumptions are satisfied.
- It is commonly used for large samples; however, it can also be used for small samples when the population standard deviation is known and the population is normally distributed.

Formula

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Where:

- $\bar{x}$ = sample mean
- μ = population mean
- σ = population standard deviation
- n = sample size

- A hospital knows that the population mean waiting time is 30 minutes, with a population standard deviation of 5 minutes. A sample of 25 patients has a mean waiting time of 32 minutes.
- Given:
- Population mean = 30
- Population standard deviation = 5
- Sample mean = 32
- Sample size = 25

Calculation:

$$Z = \frac{32 - 30}{5/\sqrt{25}}$$

$$Z = \frac{2}{1} = 2$$

The calculated Z-value is **2**.

t-test

- A t-test is a statistical test that uses the Student's t-distribution.
- We generally use a t-test when the population standard deviation sigma is unknown and we estimate it using the sample standard deviation s.
- Population standard deviation is unknown.
- Sample standard deviation is used instead.
- It is particularly useful for small samples when the population is approximately normally distributed.

Formula (one-sample t-test):

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Where:

- $\bar{x}$ = sample mean
- μ = hypothesized population mean
- s = sample standard deviation
- n = sample size

Example:

- A researcher wants to test whether the average patient waiting time differs from 30 minutes. A sample of 10 patients has a mean waiting time of 32 minutes and a sample standard deviation of 4 minutes.
- Given:
- Population mean = 30
- Sample mean = 32
- Sample standard deviation = 4
- Sample size = 10

Calculation:

$$t = \frac{32 - 30}{4/\sqrt{10}}$$

$$t = \frac{2}{1.265} \approx 1.58$$

The calculated t-value is approximately **1.58**, with $10 - 1 = 9$ degrees of freedom.

Conditions for Z-test and t-test

- **Condition 1: Sample size**
- Sample size n >=30: The sampling distribution of the mean is often approximately normal, especially when the population is not extremely skewed and has no severe outliers.
- Sample size n < 30: If the population is approximately normally distributed, we can use a t-test when the population standard deviation is unknown, or a Z-test when it is known.
- Important: A sample size below 30 does not automatically mean we must use a non-parametric test.

- *When do we use a non-parametric test?*
- A non-parametric test may be appropriate when the assumptions of a parametric test are not satisfied, depending on the research question and data type.

Examples for students

Situation	Appropriate test
Sample size 50; population SD known	Z-test, if assumptions hold
Sample size 15; population SD unknown; population approximately normal	t-test
Sample size 12; strongly skewed data with serious outliers	Consider a suitable non-parametric or robust method

What is a right-tail test

A **right-tail test** is a hypothesis test in which we want to check whether a population value is **significantly greater than a particular value**.

For example, a company claims that the average weight of a packet is 500 grams. We want to test whether the actual average weight is **greater than 500 grams**.

- $H_0 : \mu = 500$ — Null hypothesis
- $H_1 : \mu > 500$ — Alternative hypothesis

Because the alternative hypothesis uses the **greater than** ($>$) **sign**, we use a right-tail test.

What is a left-tail test?

A **left-tail test** is a hypothesis test used when we want to check whether a population mean is **significantly less than a particular value**.

For example, a company claims that the average weight of a packet is 500 grams. We want to test whether the actual average weight is **less than 500 grams**.

- Null hypothesis: $H_0 : \mu = 500$
- Alternative hypothesis: $H_1 : \mu < 500$

Because the alternative hypothesis uses the **less-than** ($<$) **sign**, we use a left-tail test.

What is a two-tail test?

A **two-tail test** is a hypothesis test used when we want to check whether a population mean is **different from a particular value**. It can be either greater than or less than that value.

For example, a company claims that the average weight of a packet is 500 grams. We want to test whether the actual average weight is different from 500 grams.

- Null hypothesis: $H_0 : \mu = 500$
- Alternative hypothesis: $H_1 : \mu \neq 500$

The symbol $\neq$ means **not equal to**.

Because the actual mean could be either greater than or less than 500, we use a two-tail test.

Skewness

- Positive skewness (Right-skewed):

  Mean>Median>Mode

- Negative skewness (Left-skewed):

  Mean<Median<Mode

- Symmetrical distribution

  Mean=Median=Mode

P201	50	82	145/95	B+	220	Yes	Low	
P202	28	58	115/75	O+	155	No	High	

3.		Three colleagues, Turjo, Arunob, and Tanvir, are collaborating on an analytical research project concerning the Rohingya issue. From the outset, they recognized that collecting comprehensive information on the Rohingya population would be challenging. Consequently, they decided to employ sampling techniques to obtain representative data. i. Turjo proposed dividing the population into five distinct age groups: 0–20, 21–40, 41–60, 61–80, and 81 and above. From each group, participants would be selected to ensure age-based representation.	[5]

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		ii. Tanvir suggested identifying the geographic locations in Bangladesh where Rohingyas reside, segmenting these areas into different regions, and then selecting participants from each region. iii. Arunob, however, emphasized that many Rohingyas may conceal their identity, making it difficult to identify them directly. He therefore recommended using a referral-based approach: beginning with Rohingyas known to the researchers, who would then introduce additional participants, thereby expanding the sample through successive referrals.	-	
	a)	**Apply** appropriate sampling techniques to collect representative data on the Rohingya population for a research study by selecting one or more of the propc sed methods (age-group sampling, geographic-region sampling, or referral-based sampling) and describing in practice how you would select participants while accounting for practical constraints such as accessibility and representativeness.		
4.		Prior to the upcoming World Cup, Bangladesh's leading batsman, Saif Hasan, had shown remarkable consistency in One Day Internationals (ODIs). In the twelve matches before the ODI series against Australia, he had never scored fewer than fifty runs. However, in Bangladesh's final series before the World Cup, his performance declined, with scores of 0, 4, and 0. The sequence of his last fifteen ODI scores is: ST, 100, 56, 59, 160, 76, 54, 100, 90, 89, 81, 98, 0, 4, 0. A Bangladeshi cricket selector believes that one or two bad innings should not overshadow his overall ability, and wants to assess his performance statistically before the World Cup.	[10]	C
	a)	Calculate the mean and median for Saif Hasan's 15 innings. **Analyze** the difference between these two measures in the context of his recent performance slump. Based on the distribution of the scores, which statistic provides the most reliable indicator of his "true" form for the World Cup selectors, and why?		
	b)	Construct a boxplot for the 15 innings. Using the Interquartile Range (IQR) method, mathematically **analyze** the dataset to determine if any of the recent low scores (0, 4, 0) qualify as formal statistical outliers. Explain how the overall skewness of the distribution either supports or contradicts the selector's belief that the recent bad innings are just an anomaly.		
	c)	The selector believes Saif's true batting average remains strong despite recent matches, claiming his true mean is above 50. A hypothesis test is conducted (H_0: $\mu = 50$ versus H_a: $\mu > 50$) at a 5% significance level, yielding a p-value of 0.03. **Analyze** this statistical result by applying the decision rule to the null hypothesis, and explain exactly what this outcome implies for the selector's confidence in picking Saif for the World Cup.		

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Hypothesis Testing

- The selector believes that Saif's true batting average remains above 50.
- Given:
- Significance level: alpha=0.05
- p-value = 0.03
- Null hypothesis: H0:mu=50
- Alternative hypothesis: H1:mu>50

- Step 1: State the hypotheses

  H0:mu=50
- The true batting average is 50.

  H1:mu>50
- The true batting average is greater than 50.

- Step 2: Decision rule
- If p-value is less than significance level, reject the null hypothesis.
- If p-value is greater than significance level, fail to reject the null hypothesis.
- Step 3: Compare p-value with significance level
- Given:

  p=0.03

  alpha=0.05
- Since:
- 0.03<0.05 . So, Reject the null hypothesis (H_0)

- Step 4: Conclusion
- There is statistically significant evidence at the 5% significance level that Saif Hasan's true batting average is greater than 50.

- One expert claims that the average oxygen intake of those who exercise regularly is higher than the average oxygen intake of older adults. Previous research has shown that the average oxygen intake in normal adults is 36.6 ml / kg. You wanted to verify the validity of this claim. That's why you randomly selected 15 people who exercised regularly and looked at the data collected from them, taking an average of 40.6 ml / kg of oxygen. The standard deviation of the sample is 8 ml / kg. It is already known that the amount of oxygen intake in adults follows the normal distribution. Now you need to verify the validity of this claim at 5% Level of Significance.

- We need to determine whether people who exercise regularly have an average oxygen intake greater than the known average of 36.6 ml/kg.
- Since the population standard deviation is unknown and the sample size is small (15), we will use a one-sample t-test (right-tailed test).

Information	Value
Sample size (n)	15
Sample mean ($\bar{x}$)	40.6 ml/kg
Population mean (μ_0)	36.6 ml/kg
Sample standard deviation (s)	8 ml/kg
Significance level (α)	5% = 0.05
Distribution	Normal

Formulate the hypotheses

- The expert claims that the average oxygen intake is higher than 36.6 ml/kg.
- Null hypothesis (H_0):

  H0:mu=36.6 , The average oxygen intake is equal to 36.6 ml/kg.
- Alternative hypothesis (H_1):

  H1:mu>36.6
- The average oxygen intake is greater than 36.6 ml/kg.
- It is a Right-tailed test, Because the expert claims that the average is higher than the population mean.

- **Select the appropriate statistical test**
- We use the t-test because:

1.Sample size is small (n = 15).

2.Population standard deviation is unknown.

3.Sample standard deviation is given (s = 8).

4.The population follows a normal distribution.

- Therefore, we use the one-sample t-test.

- Calculate the degrees of freedom:

  df=n-1 =15 -1=14
- Calculate the t-statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

n = 15

$\bar{x}$=40.6

μ= 36.7

s= 6

α=0.05

df=n-1= 15-1 =14

- t=1.936

- The decision rule is:
- If calculated t-value is greater than 1.761 → Reject H_0.
- If calculated t-value is less than or equal to 1.761 → Fail to reject H_0.
- Here,
- 1.936>1.761, so reject H0

- There is sufficient evidence at the 5% significance level to support the expert's claim that regular exercisers have a higher average oxygen intake than 36.6 ml/kg.

- The mayor of a city estimates that the average value of the assets of families living in his city is at least, 300,000. To verify the validity of this claim / assumption, he collected data from 25 families through random samples. The average wealth of the families was found to be 26,000 from the sample. Past research has shown that the population standard deviation in the distribution of wealth to the people of that city is 60,000. Now verify the veracity of the mayor's claim at 2.5% level of synapses.

Step-1: Determining Null and Alternative Hypothesis

$$H_0: \mu \geq \$300{,}000$$
$$H_1: \mu < \$300{,}000$$

Step-2: Determine which distribution to follow

As Sample size is less than 30 but standard deviation is known so we follow the Normal distribution

Step-3: Find the value of p-Value

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{288{,}000 - 300{,}000}{16{,}000} = -.75$$

p-value=0.2266

Make the decision

- Given:

alpha=0.025

- Compare:

p=0.2266

alpha=0.025

- Since:

0.2266>0.025 , Fail to reject H_0

- There is insufficient statistical evidence to conclude that the average family wealth is less than 300,000.

p-value < 0.025	Reject H_o
p-value > 0.025	Fail to reject H_o

- $Z = 0 \rightarrow$ Probability = 0.50
- $Z = -1.96 \rightarrow$ Probability = 0.025
- $Z = +1.96 \rightarrow$ Probability = 0.975
- Negative $Z \rightarrow$ Area to the left is less than 0.50.
- Positive $Z \rightarrow$ Area to the left is greater than 0.50.